The Myth of the Moravec Paradox

> Encoded in the large, highly evolved sensory and motor portions of the human brain is a billion years of experience about the nature of the world and how to survive in it. The deliberate process we call reasoning is, I believe, the thinnest veneer of human thought, effective only because it is supported by this much older and much more powerful, though usually unconscious, sensorimotor knowledge. We are all prodigious olympians in perceptual and motor areas, so good that we make the difficult look easy. Abstract thought, though, is a new trick, perhaps less than 100 thousand years old. We have not yet mastered it. It is not all that intrinsically difficult; it just seems so when we do it.
>
> Hans Moravec

The "Moravec Paradox" is not a paradox at all: it is the result of *bad philosophizing*. In other words, unexamined assumption, unconscious metaphor, and self-congratulatory conclusions have plagued this field just as they have many other fields which depend upon shallow psychologizing and naïve introspection.

Intelligence does not reach its apex and most essential expression in abstract reasoning: *a truth table* is not the heart of intelligent behavior. Proprioception, dexterity, creative manipulation, and navigation in a complexly textured real world are the essential challenges of intelligence and therefore the foundation of all other forms of intelligence, including symbolic reasoning.

But how could it be, that symbolic capacity originates in perceptual and sensorimotor tasks? How could *being-in-the-world* result in *reasoning-about-the-world*? What's the probable evolutionary genealogy?

Let's consult ethology for a moment and consider the example of the "supernormal sign stimulus". As Nils Tinbergen famously demonstrated, an animal is not only fooled by a substitute sign, but will respond proportionally to a sign's *exaggerated* character: the baby seagull will respond all the more vigorously to the minimal but exaggerated sign stimulus of an unnaturally bright red dot on the beak of its feeding parent.

What can this tell us? That the activating gestalt for an instinctual behavior can be isolated and toyed with, because instinctual discharge is *already inherently symbolic*. As the immensely influential but still largely unknown Jakob von Uexküll convincingly showed, every animal lives within its own *Merkwelt*: there is no "raw" data, but a curved field of vectors and weighted signs.

> The animal's own world is not only dependent on what its sense organs can or cannot receive. Its sensory world is still more restricted; it is composed of sign stimuli [...]
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> N. Tinbergen, *Study of Instinct*, p.37

These ideas form the basis of the field known as "biosemiotics", which Rodney Brooks was not entirely unaware of. What situated AI attempts to prove in the shape of a functional robot, is the realization that the "sensorium" of any given intelligence is neither a *tabula rasa* as the Cartesian-Kantian psychology wants it, nor solely composed of "always-already" signs as the French structuralists want it, but the undulating inner surface of *a sphere of potential signs*. Sense data is a heavily weighted graph, with strong attractors and repulsors: to "sense" is to traverse a field of virtual signs which actuate in degrees. Every creature is both constantly hallucinating and overreacting, and constantly ignoring and remaining passive: the question is, what matters to *this* creature, at *this* time, in *this* place.

To build a viable replicant of intelligent behavior requires that we internalize this lesson and allow the symbolic reasoning to percolate from the *lowest possible levels*: the more a symbol is defined in terms of *recent* and *accurate* sensorimotor feedback, the more meaningful it is. In other words, abstraction is obtained with a proportional sacrifice of relevance: most of the apparently symbolic behavior of a living creature, is merely imputed.